Straight-Line Graphs

Answer sheet · 26 marks total

Section A — Reading gradient and y-intercept

A1. **6** (1)   A2. –**4** (1)   A3. –**2** (1)   A4. **9** (1)

Section B — Finding the gradient from two points

B1. (10–2) ÷ (3–1) (1)   = 8÷2 = **4** (1)

B2. (–1–7) ÷ (4–0) (1)   = –8÷4 = –**2** (1)

B3. (4–(–4)) ÷ (1–(–3)) (1)   = 8÷4 = **2** (1)

Section C — Tables and equations from data

C1. y-values: –5, –2, 1, 4 (1)   points: (–1,–5), (0,–2), (1,1), (2,4) (1)   all lie on a straight line (1)

C2. gradient = (5–(–1)) ÷ (3–0) = 6÷3 = 2 (1)   y-intercept c = –1 (from x=0) (1)   = **y = 2x** – **1** (1)

Section D — Parallel lines and forming equations

D1. parallel → gradient = 4 (1)   using (0,6): c = 6 → **y = 4x + 6** (1)

D2. both lines have gradient 3 (1)   so **yes**, they are parallel (1)

D3. gradient = (13–5) ÷ (6–2) = 8÷4 = 2 (1)   substitute (2,5): 5 = 2(2) + c (1)   5 = 4 + c → c = 1 (1)   = **y = 2x + 1** (1)

D4. substitute x = 3: y = 2(3) + 1 = 7 (1)   matches the given y-value, so **yes**, the point lies on the line (1)

Marking note: award method marks for a correctly substituted gradient formula or correctly identified gradient/intercept, even if the final answer contains an arithmetic error.